FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 2

B.Sc. MATHEMATICS HONOURS

MAT5CJ303: COMPLEX ANALYSIS I

Module II: Special Power Functions (z^n, $z^{1/n}$, and Principal nth Root Functions)

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Define a complex polynomial function and a complex power function. Explain geometrically how the complex squaring mapping $w = z^2$ acts on a nonzero point $z = re^{i\theta}$ in terms of magnification and rotation.	**3**
2	Find the image of the circular arc defined by $\lvert z\rvert = \frac{1}{2}$, $0 \le \arg(z) \le \pi$, under the complex mapping $w = z^2$.	**3**
3	Find the image of the ray $\arg(z) = \frac{\pi}{6}$ under each of the following complex power mappings: (a) $f(z) = z^3$, (b) $f(z) = z^4$, and (c) $f(z) = z^5$.	**3**
4	Define a one-to-one complex function and its inverse function f^{-1}. Show that the complex function $f(z) = z + 3i$ is one-to-one on the entire complex plane and find a formula for $f^{-1}(z)$.	**3**
5	Define the principal square root function $z^{1/2}$. Using this definition, find the values of $z^{1/2}$ at the points: (a) $z = -2i$, and (b) $z = -1 + i$.	**3**
6	Find the image of the circular arc $\lvert z\rvert = 9$, $-\frac{\pi}{2} \le \arg(z) \le \pi$, under the principal square root mapping $w = z^{1/2}$.	**3**
7	If z is a nonzero complex number, write the exponential and polar expressions for the principal n th root function $z^{1/n}$ ($n \ge 2$). State its natural domain and range in the complex plane.	**3**
8	Find the value of the given principal nth root function at the indicated point: (a) $z^{1/3}$ at $z = i$, and (b) $z^{1/4}$ at $z = -1 + \sqrt{3}i$.	**3**
9	What is meant by a multiple-valued function in complex analysis? Distinguish between the single-valued function $g(z) = z^{1/3}$ and the multiple-valued function $G(z) = z^{1/3}$ by evaluating both at $z = i$.	**3**
10	Define a rational power function $z^{m/n}$ as a composition of the principal nth root function and the power function z^m. Using this definition, verify that $(8i)^{2/3} = 2 + 2\sqrt{3}i$.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Find the image of the vertical line $x = k$ ($k \ne 0$) under the mapping $w = z^2$. Hence, describe the images of the pair of vertical lines $x = \pm 3$ and the imaginary axis $x = 0$ under $w = z^2$.	**6**
12	Determine the image of the quarter disk defined by the inequalities $\lvert z\rvert \le 2$, $0 \le \arg(z) \le \frac{\pi}{2}$, under the mapping $w = z^3$. Also, find three distinct sets in the complex plane that map onto the ray $\arg(w) = \pi$ under $w = z^3$.	**6**
13	Find the image of the annular sector $1 \le \lvert z\rvert \le 2$, $\frac{\pi}{4} \le \arg(z) \le \frac{3\pi}{4}$, under each of the following composite mappings: (a) $f(z) = 3z^2 + i$, and (b) $f(z) = \frac{1}{2}z^4 - i$.	**6**
14	Show that the function $f(z) = z^2$ is one-to-one on the restricted domain A defined by $-\frac{\pi}{2} < \arg(z) \le \frac{\pi}{2}$. Explain why $f(z) = z^n$ ($n \ge 2$) is not one-to-one on $\mathbb{C}$ by exhibiting n distinct points that map onto the same point $w = r^n e^{in\theta}$.	**6**
15	Find the image of the circular sector S defined by $\lvert z\rvert \le 3$, $\frac{\pi}{2} \le \arg(z) \le \frac{3\pi}{4}$, under the principal square root function $w = z^{1/2}$. Also, find the image of the parabola $x = \frac{9}{4} - \frac{y^2}{9}$ under $w = z^{1/2}$.	**6**
16	Let θ be the unique value of the argument of z in the interval $\pi < \theta \le 3\pi$. Show that the square root function defined by $f(z) = \sqrt{\lvert z\rvert}e^{i\theta/2}$ is the negative of the principal square root function $z^{1/2}$, and evaluate $f(i)$.	**6**
17	Using the definition of the principal nth root function $z^{1/n} = \sqrt[n]{\lvert z\rvert}e^{i\mathrm{Arg}(z)/n}$, evaluate the following: (a) $z^{1/3}$ for $z = -3 + 3i$, (b) $z^{1/5}$ for $z = 1 - \sqrt{3}i$, and (c) $z^{1/5}$ for $z = -4\sqrt{3} + 4i$.	**6**
18	Show that the complex power function $f(z) = z^n$ ($n > 2$) is one-to-one on the restricted domain $-\frac{\pi}{n} < \arg(z) \le \frac{\pi}{n}$ ($z \ne 0$), and deduce the algebraic formula, domain, and range of its inverse function, the principal nth root function $z^{1/n}$.	**6**
	Section C	

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	(a) Find the image of the triangle with vertices 0, $1+i$, and $1-i$ under the mapping $w=z^2$, and represent the mapping by describing the image of each of the three sides. (b) Use the real and imaginary parts of $w=z^2$ to find the images of the hyperbolas $xy=k$ ($k\neq 0$) and $x^2-y^2=k$ ($k\neq 0$) under the mapping $w=z^2$.	**10**
20	(a) Prove that the inverse function of $f(z)=z^2$ defined on the restricted domain $-\frac{\pi}{2}<\arg(z)\leq\frac{\pi}{2}$ ($z\neq 0$) is the principal square root function $z^{1/2}=\sqrt{\lvert z\rvert}e^{i\operatorname{Arg}(z)/2}$. Compare the mapping behavior of $w=z^2$ and $w=z^{1/2}$ on the rays $\arg(z)=\pm\frac{\pi}{2}$ and the negative real axis. (b) Discuss the construction of the Riemann surface for the complex squaring function $f(z)=z^2$ on the closed unit disk $\lvert z\rvert\leq 1$, and explain how it helps visualize the multiple-valued function $F(z)=z^{1/2}$.	**10**